

Rational Expressions & Equations

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

☐ 1. I can **state** excluded values for a rational expression.

RELATED SKILLS

☐ I Determine excluded values from an original denominator.☐ I Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.

☐ 2. I can **simplify** a rational expression while preserving its restrictions.

RELATED SKILLS

☐ I Cancel common factors rather than individual terms.☐ I Factor numerators and denominators completely.☐ A Determine excluded values from an original denominator.☐ A Recall and correctly use rational-expression terms including numerator, denominator, restriction, and excluded value.

☐ 3. I can **multiply and divide** rational expressions.

RELATED SKILLS

☐ A Add, subtract, multiply, and divide fractions.☐ A Cancel common factors rather than individual terms.☐ A Determine excluded values from an original denominator.☐ A Factor numerators and denominators completely.

☐ 4. I can **add and subtract** rational expressions using a common denominator.

RELATED SKILLS

☐ I Combine rational expressions over a common denominator.☐ I Determine the least common denominator of rational expressions.☐ A Add, subtract, multiply, and divide fractions.

☐ 5. I can **solve** a rational equation and reject extraneous solutions.

RELATED SKILLS

☐ I Check candidate solutions against denominator restrictions.☐ I Clear denominators from a rational equation.☐ A Determine excluded values from an original denominator.☐ A Use inverse operations to maintain an equivalent equation.